

Why the market wins

Four strategies for prediction and crypto markets, tested against live data and falsified

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Over one week, four plausible ways to make money on Polymarket and short-horizon crypto markets were specified in advance, instrumented, and measured against live order books and on-chain data. All four failed. This document records what was measured, how, and why the failures were predictable in hindsight.

The central finding is not that these particular strategies do not work. It is that five separate measurement errors each produced a convincing false edge before being caught. Anyone acting on the first reading of any of them would have lost money with full confidence.

All four falsified — by measurement, not by opinion

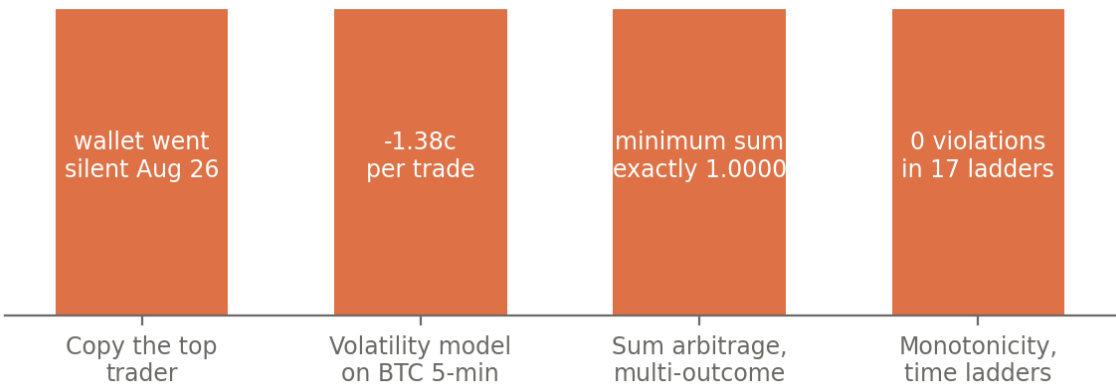


Figure 1. The four hypotheses and the evidence that ended each.

Summary of findings

Hypothesis	Method	Result
Mirror the trades of the platform's most profitable wallet	On-chain activity API, 39,408 trades reconstructed	Not replicable. 15,700 buys/day across 3,491 markets, in slices of ~120 per market. Wallet stopped trading 26 Aug 2026.
Trade Polymarket's 5-minute crypto contracts against a volatility model	46 h live collection, 72,832 quotes, 4,427 markets, resolutions validated to 99.2%	-1.38 cents per trade over 3,745 trades. Market well calibrated in all ten price bands.
Buy every outcome of a multi-outcome event for less than \$1	Order books for 949 outcomes across 142 events	Impossible. Minimum observed sum was exactly 1.0000; the platform enforces it.
Exploit logical inconsistency in cumulative 'by date X' ladders	17 ladders, 260 order books, monotonicity checked pairwise	Zero violations.

A note on what 'falsified' means here

None of these results prove that no edge exists anywhere in these markets. They establish something narrower and more useful: each specific, pre-registered strategy was measured against real executable prices and real outcomes, and none produced a return that survives the cost of crossing the spread. Where a confidence interval is reported, it is stated rather than rounded toward the conclusion.

Hypothesis 1: copy the best trader

The starting point was a widely shared claim: a Polymarket account had turned \$111 into \$4.3 million in seven months. The account's lifetime profit later reached \$23.6 million. The proposed strategy was to mirror its positions at a fixed stake of 10 EUR.

Reconstructing the wallet's activity from the on-chain API produced 39,408 trades covering a 60-hour window. The pattern rules out mirroring:

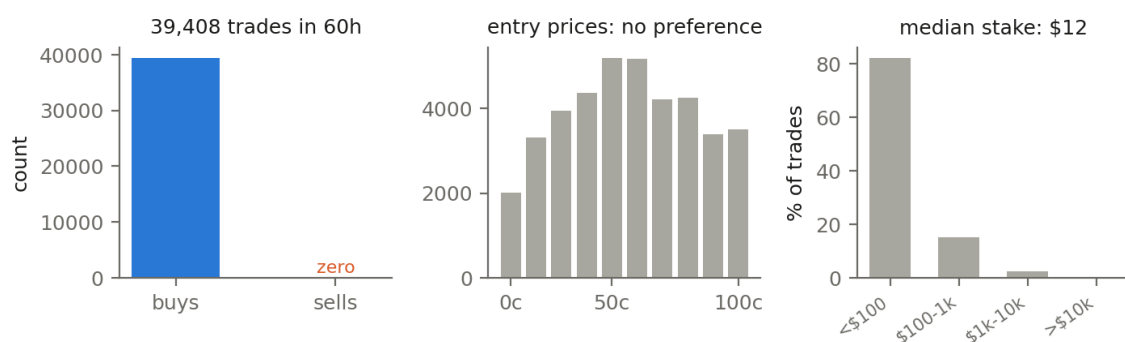


Figure 2. Left: the wallet bought 39,408 times and sold zero times, holding every position to settlement. Centre: entry prices are spread almost uniformly from 1 to 99 cents, so there is no preferred price region. Right: 82% of trades were under \$100, with a median of \$12.

The decisive number is not the size but the granularity. The wallet touched 3,491 distinct markets in 2.5 days, entering a single football match up to 217 times in separate slices. There is no discrete 'bet' to copy: a follower mirroring the first slice buys near the best available price, while one mirroring the eightieth buys after the wallet has already moved the book against them.

Separately, the account's last recorded trade was an \$8 position on 26 August 2026. Its open position value had fallen from \$426,900 to \$27,100, a 94% reduction. The viral post describing the \$4.3 million was already seven months stale when it was read.

Where the framing misleads. A headline figure of lifetime profit is not a track record. Over a separate 90-day window the same wallet showed a negative realised result. Copying a strategy from the moment it becomes famous means buying the part that came after the run, not the run.

Hypothesis 2: a volatility model on 5-minute crypto contracts

Polymarket lists contracts resolving every five minutes on whether a cryptocurrency finishes higher than it started. These settle on a Chainlink 60-second time-weighted average price. Because the strike, the spot price and the time remaining are all known exactly, the fair price of such a contract depends on exactly one unknown quantity: volatility.

That reduces the entire question to a single testable claim. If the volatility the market prices in differs systematically from the volatility the asset actually exhibits, the difference is tradable. No view on direction is required at any point.

Method

A collector recorded live order books for seven cryptocurrencies every 15 seconds for 46 hours, alongside one-second price data from a spot exchange. For each observation it computed the implied volatility from the market price and the realised volatility from actual price movement. Resolutions were reconstructed independently and validated at 99.2% agreement against prices recorded seconds before each window closed.

A trading rule was fixed in code before any results were examined: buy when the model price beats the executable price by more than 3 cents, only between 60 and 240 seconds before settlement, one trade per market. This pre-registration matters. Without it, the thresholds could be adjusted after seeing the data until the result looked favourable.

Data collected	Value
Collection window	46.0 hours, 0.4 hours lost to interruptions
Quotes recorded	72,832, all from live order books
Markets resolved	4,427
Median bid-ask spread	1.00 cent
Strike-derivation error at window open	-0.18 basis points (median)

Result: the market is well calibrated

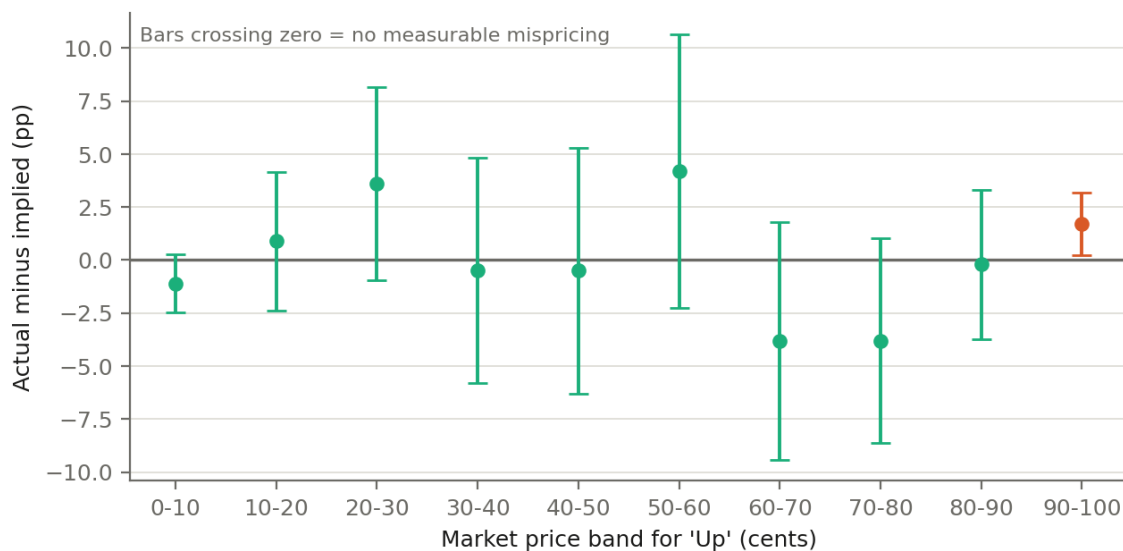


Figure 3. For each price band, the difference between how often the event actually occurred and how often the price implied it would. Error bars are 95% intervals. Nine of ten bands cross zero. The one marginal exception is what ten simultaneous tests produce by chance.

Contracts priced at 25 cents resolved favourably about a quarter of the time. There is no systematic bias toward longshots, which was the most plausible candidate before the data arrived, and no pattern in the direction of the deviations.

The pre-registered rule lost money:

Metric	Value
Trades triggered	3,745 (of 4,427 markets)
Hit rate	43.2%
Mean result	-1.38 cents per trade
95% interval (block bootstrap)	-3.96 to +1.30 cents
At a 10 EUR stake	-516 EUR over 46 hours

Stated precisely: the interval includes zero, so the rule is not proven to lose. What is established is that it shows no profit, and the independent calibration result points the same way.

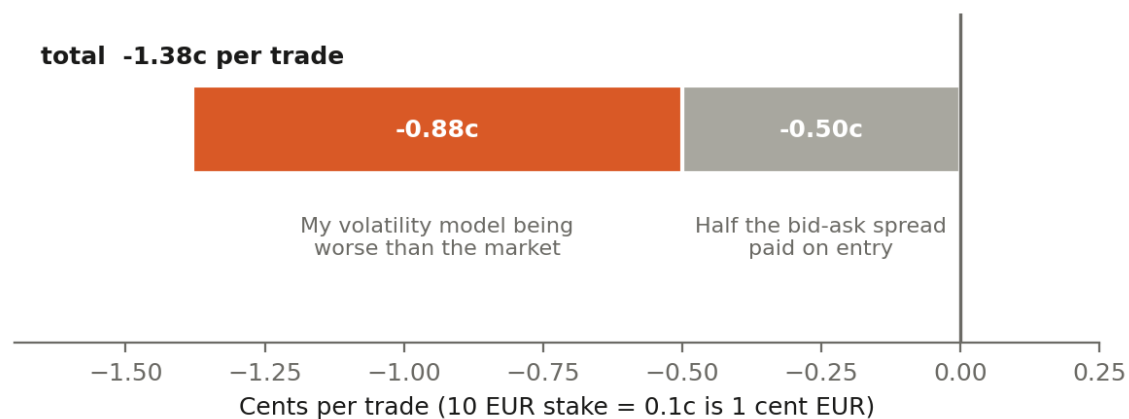


Figure 4. The loss splits cleanly. Half a cent is the unavoidable cost of buying at the ask rather than the midpoint. The remainder is the model being a worse estimator of volatility than the market itself.

The most informative statistic is one that was almost overlooked. The rule fired on 85% of all markets. A filter that triggers five times in six is not selecting anything; it is reporting that the model disagreed with the market almost everywhere. That is the signature of a noisy estimator, not of a mispriced market.

This is a structural point rather than a flaw in one implementation. Bitcoin volatility is continuously traded on liquid options venues by firms whose entire business is estimating it. A fifteen-minute realised volatility estimate is not going to beat that consensus.

Hypotheses 3 and 4: arithmetic, not modelling

After the model-based approach failed, the remaining candidates were chosen precisely because they require no model at all. If several contracts are logically linked, arithmetic alone determines whether their prices are consistent. One cannot be wrong about arithmetic.

Sum arbitrage. When exactly one outcome of an event can occur, buying all outcomes for less than \$1 guarantees a profit. Across 142 events and 949 order books, the minimum sum observed was exactly 1.0000. Polymarket's negative-risk mechanism enforces this at the platform level.

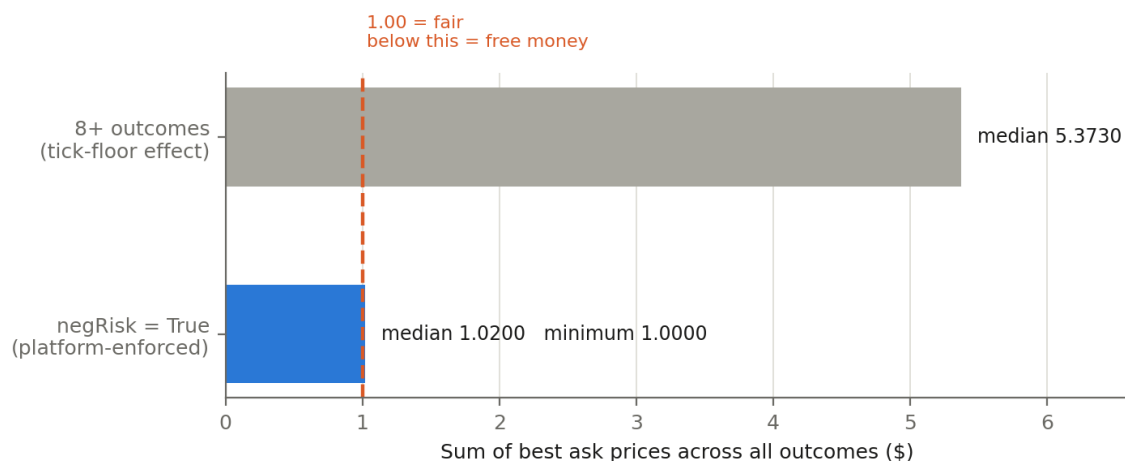


Figure 5. Sums of best ask prices. The platform holds enforced markets just above the fair value of 1.00. Separately, a minimum tick of 0.001 means an event with many outcomes cannot sum below 1 even in principle: a hundred hopeless candidates cost 10 cents in tick floor alone.

Monotonicity in time ladders. Markets asking whether an event will occur by successive dates are cumulative: anything true by September is true by December, so the price must rise with the date. Seventeen such ladders were checked pairwise across 260 order books. Every sequence was monotonic. Zero violations.

The part that mattered: five false edges

Each of the following produced a clear, plausible, and entirely spurious signal. Each was caught by a check rather than by intuition. Acting on any of them would have meant trading confidently in the wrong direction.

What appeared	Apparent edge	Actual cause
Market priced 'Up' at 49.5% while spot sat above the strike	12 percentage points	The strike was off by 2.3 basis points. At a 4-minute horizon a rounding difference moves the implied probability by ten points.
Model and market diverged more and more within each window	up to 50 points	Prices were read from a cached API field lagging by minutes. The live book said 0.965 while the cache said 0.455.

What appeared	Apparent edge	Actual cause
Implied volatility of 120,850% annualised	absurd	Measured 2 seconds before settlement, where the effective time term approaches zero and the formula degenerates.
Resolution lookups returned nothing at all	n/a	Resolved markets disappear from the slug endpoint. Ground truth was recovered from prices already recorded seconds before each close.
Trades looked strongly clustered in time, implying overstated significance	unknown	Tested across 80 simulated runs. Both naive and block-bootstrap intervals held their 5% false-alarm rate. The concern was real but immaterial here.

The last row is included deliberately. It was a suspicion that sounded correct, was tested, and turned out not to matter. Testing a concern rather than asserting it is the same discipline that caught the other four.

The general lesson is sharper than any of the individual bugs. At five-minute horizons the mapping from price to probability is extremely sensitive: an error small enough to be ignored in almost any other context shifts the implied probability by ten points or more. In such a setting, almost every mistake first presents itself as an opportunity.

Working rule. When a measurement shows a large edge, the first question is not how to trade it. It is what is broken. Four times out of five during this project, something was.

Why the market wins

A price is not an opinion. It is the residue of every participant with capital, computing power and full-time attention who has already asked the same question. What survives that process is what is not worth their time.

Every approach attempted here fell into one of two categories. The model-based approach required estimating something that specialist firms estimate better. The arithmetic approaches required finding an inconsistency that a program can detect, which means a program already has. Both categories are closed by construction, not by accident.

What remains open is the work that does not scale: reading resolution criteria that most participants skim, and holding genuine domain knowledge in a niche too small for institutional capital. Both resist automation, which is exactly why they persist. Neither was available to this project.

On position size. At a 10 EUR stake, a genuine 3% edge is worth 30 cents per trade against a per-trade standard deviation exceeding 40 cents. Thousands of trades would be required to distinguish skill from luck. Below a certain capital base, returns are simply the wrong lever: the arithmetic favours increasing contributions over increasing yield, by a wide margin.

What was built

The measurement apparatus outlasts the negative results: a collector pairing live order books with one-second exchange data; a resolution reconstruction validated at 99.2% against recorded ground truth; an evaluation pipeline verified in both directions, detecting a planted edge and reporting none on a fair market; and a scanner for logical inconsistencies across linked contracts.

Four hypotheses were specified in advance, instrumented, measured and abandoned on the evidence. The total cost was zero. The same conclusions purchased through live trading would have cost 516 EUR in the second case alone, and would have arrived later and less clearly.